

SCIENCE IS THE PATH TO KNOWLEDGE, AND FINDING OUT ABOUT HOW OUR UNIVERSE WORKS. YOU CAN'T BE A GEEK AND NOT KNOW YOUR SCIENCE!

THIS MONTH IN SCIENCE:

The rare emerald icebergs, the mystery of the Kuiper Cliff, insulin pills and tractor beams...



Lightning strikes twice!

A study has revealed that lightning often strikes twice, contrary to popular belief. A lightning strike occurs when a negative charge buildup in a thundercloud finds a channel to the ground through ionised air. The charge is not drained completely in a single strike, and the subsequent release can happen through the same channel.

<https://dgit.in/litstrik>

WHAT'S NEW

The impact of flying cars

A first of its kind study investigating the environmental impact of vertical takeoff and landing aircraft (VTOLs), colloquially known as flying cars, has found out that they would not offer a more sustainable option over short distances. The study examined the greenhouse gas emissions, energy use, and time saved as compared to the equivalent trips on conventional ground-based vehicles. For trips of 100 kilometers, ferrying three passengers, the flying cars would provide a more environmentally friendly option. While the vehicle itself would not have any emissions, there are emissions tied to the generation of the required electricity at power plants. In all,



the trip would be associated with 52 percent lower emissions than one made with a petrol based vehicle, and 6

percent lower than an electric vehicle. However, for trips shorter than 35 kilometers, even with single occupancy, the conventional vehicles had a lower footprint. The average commute distance is 17 kilometers. The finding indicates that a world like the one shown in the *Jetson's* would not be very good for the environment.

<https://dgit.in/jetcar>

The day the dinosaurs died

Around 65 million years Ago, the Chicxulub meteorite impacted at the Yucatan peninsula, and is believed to be related to one of the largest mass extinctions known. The impact caused seismic waves, and associated surges that moved around the world faster than tsunamis could. Evidence of such surges is only known from marine deposits, and were unknown in the continental record... so far. Now, a



newly discovered site in the Hell Creek Formation of North Dakota, is a continental record of a high energy surge. This is an exquisitely preserved site, and has an ultra high temporal fidelity. It is the first site to

provide a complete record of the Chicxulub impact event in a sub twenty four hour scale, and serves as a diary in stone. There are a variety of organisms fossilised in the record, including reptiles, dinosaurs, ammonites, fish, shark, micro-fauna, branches and trees. The site provides a record of the last day of the dinosaurs, and the studies on the site are only just beginning.

<https://dgit.in/dinoded>



Earliest life emerged in ponds?

Shallow pools of water less than 10 cm deep could trap enough nitrogen to catalyse the earliest forms of life. This key ingredient would not have been available in large enough quantities in oceans. A new study indicates that the earliest forms of life on Earth in ponds, not oceans.

<https://dgit.in/pndlyf>



3D printed organs and food

Additive manufacturing of tissue has been difficult because of the time needed to print anything large enough to be useful, due to how quickly the tissue deteriorated. A new device uses multiple printers to manufacture many layers of the tissue simultaneously.

<https://dgit.in/3dfood>



Fully automated DNA storage

Researchers from the University of Washington and Microsoft have successfully demonstrated fully automated DNA storage for the first time. This is a crucial step in moving the technology out of labs and making it more widely available.

<https://dgit.in/hellodna>